



DATA MODELLING - TIME SERIES ASSIGNMENT

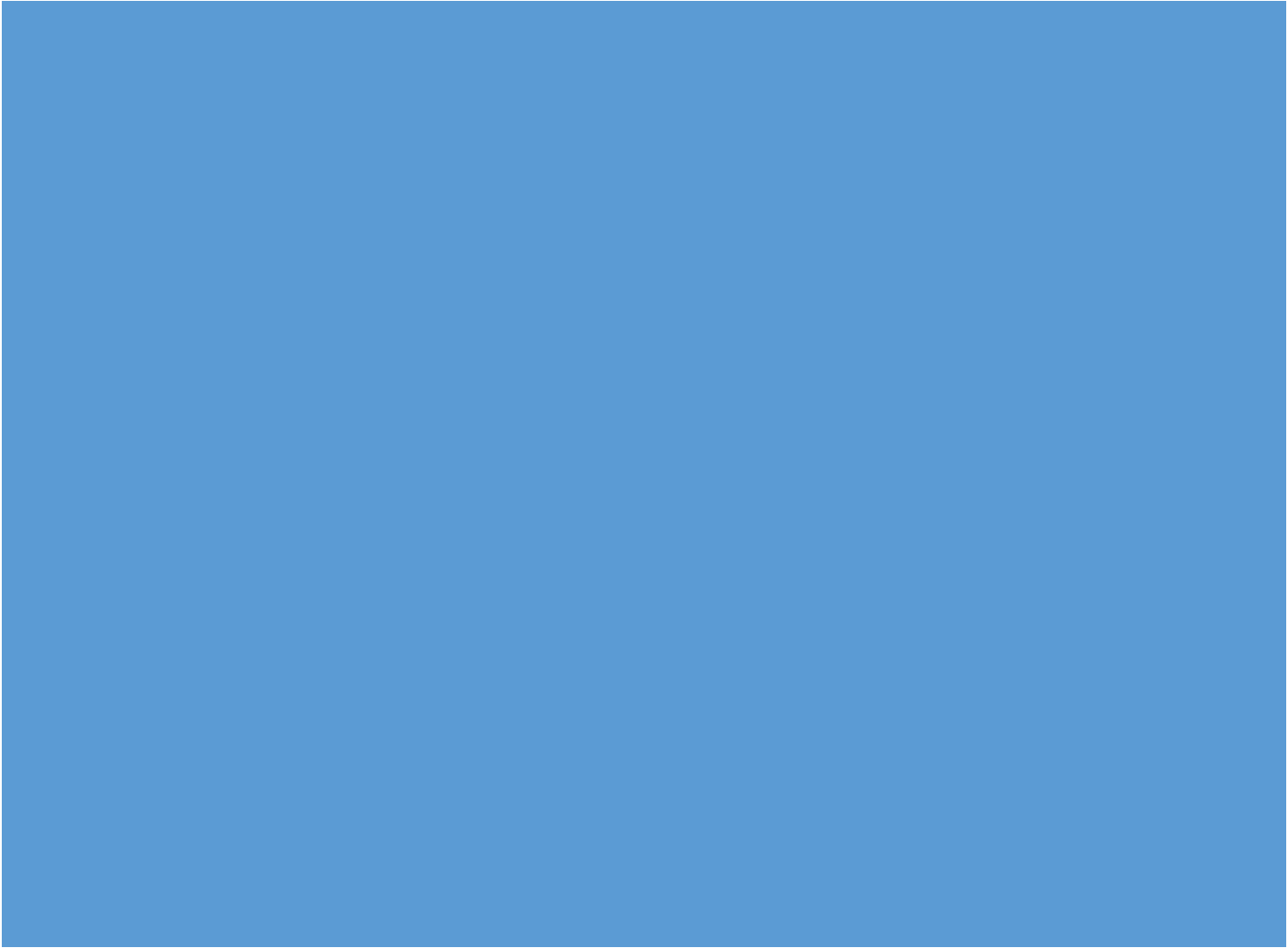


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Question 1 – Time Series Decomposition and Forecasting

1. Introduction

This report analyses the time series associated with Student ID 2523258 (Series ID 22), comprising 36 monthly observations over three years. The objective is to explore the structure of this series, identify its key components, and produce forecasts for future periods.

The analysis focuses on detecting the trend and seasonal patterns, selecting an appropriate decomposition model, and using it to forecast the next eight periods (Periods 37–44). By decomposing the series, we aim to understand the underlying behaviour and dynamics of the data.

2. Time Series Plot and Preliminary Analysis

The time series plot was generated using R software based on the 36 monthly observations.

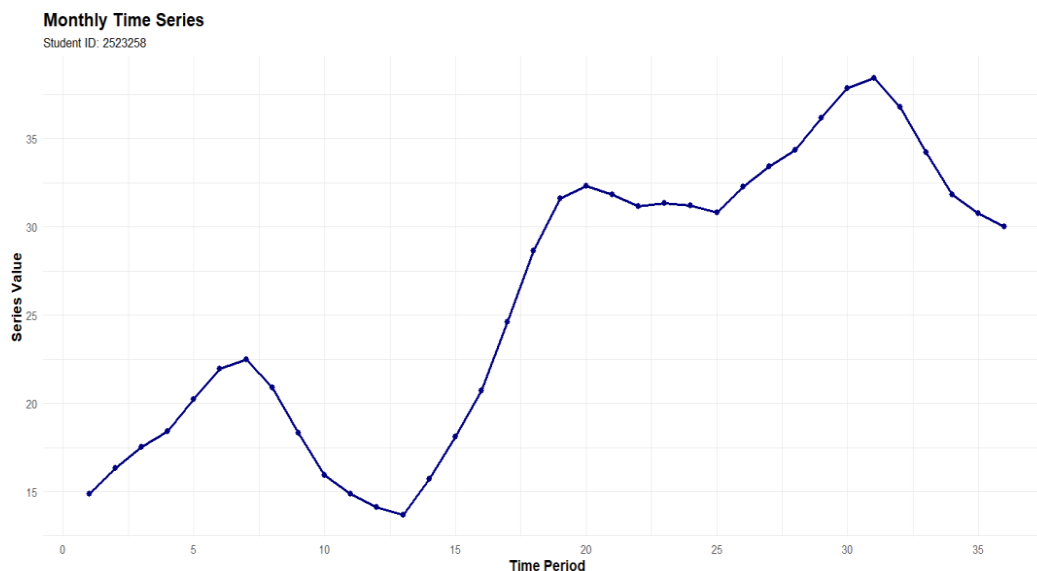


Figure 1: Time Series Plot

From the plot, several characteristics can be observed:

- An upward trend from approximately 14.889 in period 1 to above 38 in period 31, with a slight decline toward the end.
- A repeating seasonal pattern can be observed roughly every 12 months, indicating the presence of seasonality.
- Proportional fluctuations that increase with the series' level.

These characteristics indicate that the series consists of trend and seasonal components, making decomposition an appropriate analysis method.

3. Choice of Decomposition Model

Visual inspection suggests a multiplicative decomposition model is most suitable.

The model assumes:

$$Y_t = T_t \times S_t \times E_t \quad (1)$$

Where

Y_t represents the observed value at time t ,
 T_t represents the trend component,
 S_t represents the seasonal component, and
 E_t represents the irregular component.

This approach is appropriate because seasonal fluctuations vary proportionally with the series' level.

4. Estimation of the Trend Component

The trend component represents the long-term movement of a time series, excluding seasonal fluctuations. For this series, a 12-month moving average (MA) is used because the data is monthly with a 12-period seasonal cycle.

4.1 Methodology

Step 1: Compute 12-Month Moving Average (MA)

The moving average smooths short-term variations and highlights the trend:

$$MA_t = \frac{Y_t + Y_{t-1} + Y_{t-2} + Y_{t-3} + Y_{t-4} + Y_{t-5} + Y_{t-6} + Y_{t-7} + Y_{t-8} + Y_{t-9} + Y_{t-10} + Y_{t-11}}{12} \quad (2)$$

Where:

- Y_t = observed value at time t
- MA_t = 12-month moving average at time t

Step 2: Centered Moving Average (CMA)

Since a 12-month MA is even, the average is centered between two periods. The centered moving average (CMA) is calculated as:

$$CMA_t = \frac{MA_t + MA_{t+1}}{2} \quad (3)$$

CMA aligns the trend estimate with the original period, representing the trend-cycle component.

4.2 MA Table (Periods 6–30)

Table 1: Moving Average (MA) Table for periods 6-30

Period	MA (12-month)
6	17.996
7	17.894
8	17.842
9	17.890
10	18.079
11	18.442
12	18.998
13	19.756
14	20.709
15	21.837
16	23.108
17	24.482
18	25.909
19	27.337
20	28.715
21	29.994
22	31.131
23	32.095
24	32.865
25	33.433
26	33.807
27	34.005
28	34.060
29	34.013
30	33.913

Note:

1. The complete step-by-step hand calculations for all the moving averages (MA) are included in the Annex for reference.

- Periods **1–5 and 31–36** do not have 12-month moving averages because the calculation requires 12 consecutive observations. At the beginning of the series (periods 1–5), there are not enough previous data points to form the moving average. Similarly, at the end (periods 31–36), there are not enough subsequent observations to complete the 12-month window required for the calculation.

4.3 CMA Table (Periods 6–29)

Table 2: Centered Moving Average (CMA) Calculation Table for periods 6-29

Period	MA(t)	MA(t+1)	CMA calculation	CMA
6	17.996	17.894	$(17.996+17.894)/2$	17.945
7	17.894	17.842	$(17.894+17.842)/2$	17.868
8	17.842	17.890	$(17.842+17.890)/2$	17.866
9	17.890	18.079	$(17.890+18.079)/2$	17.9845
10	18.079	18.442	$(18.079+18.442)/2$	18.2605
11	18.442	18.998	$(18.442+18.998)/2$	18.720
12	18.998	19.756	$(18.998+19.756)/2$	19.377
13	19.756	20.709	$(19.756+20.709)/2$	20.2325
14	20.709	21.837	$(20.709+21.837)/2$	21.273
15	21.837	23.108	$(21.837+23.108)/2$	22.4725
16	23.108	24.482	$(23.108+24.482)/2$	23.795
17	24.482	25.909	$(24.482+25.909)/2$	25.1955
18	25.909	27.337	$(25.909+27.337)/2$	26.623
19	27.337	28.715	$(27.337+28.715)/2$	28.026
20	28.715	29.994	$(28.715+29.994)/2$	29.3545
21	29.994	31.131	$(29.994+31.131)/2$	30.5625
22	31.131	32.095	$(31.131+32.095)/2$	31.613
23	32.095	32.865	$(32.095+32.865)/2$	32.480
24	32.865	33.433	$(32.865+33.433)/2$	33.149
25	33.433	33.807	$(33.433+33.807)/2$	33.620
26	33.807	34.005	$(33.807+34.005)/2$	33.906
27	34.005	34.060	$(34.005+34.060)/2$	34.0325
28	34.060	34.013	$(34.060+34.013)/2$	34.0365
29	34.013	33.913	$(34.013+33.913)/2$	33.963

Note:

- CMA represents the trend-cycle component of the series.
- There is no MA for period 31, so CMA stops at period 29.
- These CMA values will be used in the deseasonalisation step.

5. Estimation of Seasonal Index

Formula (Multiplicative Model):

$$S_t = \frac{Y_t}{CMA_t} \quad (4)$$

Where:

- Y_t = observed series value at period t
- CMA_t = Centered Moving Average
- S_t = seasonal ratio (the relative size of the observation compared to the trend)

We will calculate ratios for periods 6–29 because CMA is available only for these periods.

5.1 Calculate Seasonal Ratios (S_t)

Table 3: Seasonal Ratios (S_t) Calculation Table

Period	Observed value (Y_t)	CMA_t	Calculation (Y_t/CMA_t)	Seasonal Ratio S_t
6	21.944	17.945	21.944 / 17.945	1.22284759
7	22.499	17.868	22.499 / 17.868	1.25917842
8	20.894	17.866	20.894 / 17.866	1.169483936
9	18.308	17.9845	18.308 / 17.9845	1.017987712
10	15.913	18.2605	15.913 / 18.2605	0.871443827
11	14.865	18.720	14.865 / 18.720	0.794070513
12	14.107	19.377	14.107 / 19.377	0.728028075
13	13.662	20.2325	13.662 / 20.2325	0.675250216
14	15.728	21.273	15.728 / 21.273	0.739340949
15	18.086	22.4725	18.086 / 22.4725	0.804805874
16	20.696	23.795	20.696 / 23.795	0.869762555
17	24.598	25.1955	24.598 / 25.1955	0.976285448
18	28.624	26.623	28.624 / 26.623	1.075160575
19	31.594	28.026	31.594 / 28.026	1.127310355
20	32.327	29.3545	32.327 / 29.3545	1.101262157
21	31.843	30.5625	31.843 / 30.5625	1.041897751
22	31.169	31.613	31.169 / 31.613	0.985955145
23	31.346	32.480	31.346 / 32.480	0.965086207
24	31.231	33.149	31.231 / 33.149	0.942140034
25	30.805	33.620	30.805 / 33.620	0.916270077
26	32.264	33.906	32.264 / 33.906	0.951571993
27	33.430	34.0325	33.430 / 34.0325	0.982296334
28	34.344	34.0365	34.344 / 34.0365	1.009034419
29	36.164	33.963	36.164 / 33.963	1.064805818

Note: Seasonal ratios greater than 1 indicate periods above the trend, while values below 1 indicate periods below the trend.

5.2 Group by Month (Seasonal Cycle)

Assuming your periods correspond to monthly data, we map periods 6–29 to months cyclically (Jan–Dec):

Table 4: Mapping of periods 6–29 to calendar months (January–December) assuming monthly periodicity.

Period	Month
6	Jun
7	Jul
8	Aug
9	Sep
10	Oct
11	Nov
12	Dec
13	Jan
14	Feb
15	Mar
16	Apr
17	May
18	Jun
19	Jul
20	Aug
21	Sep
22	Oct
23	Nov
24	Dec
25	Jan
26	Feb
27	Mar
28	Apr
29	May

5.3 Compute Average Seasonal Index per Month

The Average Seasonal Index is calculated as the mean of the seasonal ratios for all periods corresponding to that month:

$$\text{Average Seasonal Index} = \frac{S_t(\text{period1}) + S_t(\text{period2})}{2} \quad (5)$$

Table 5: Seasonal Indices Calculation Table

Month	Period (t)	Seasonal Ratio (S _t)	Calculation of Average Seasonal Index	Average Seasonal Index
Jan	13	0.675250216		
	25	0.916270077	(0.675250216 + 0.916270077) / 2	0.796
Feb	14	0.739340949		
	26	0.951571993	(0.739340949 + 0.951571993) / 2	0.846
Mar	15	0.804805874		
	27	0.982296334	(0.804805874 + 0.982296334) / 2	0.894
Apr	16	0.869762555		
	28	1.009034419	(0.869762555 + 1.009034419) / 2	0.939
May	17	0.976285448		
	29	1.064805818	(0.976285448 + 1.064805818) / 2	1.021
Jun	6	1.22284759		
	18	1.075160575	(1.22284759 + 1.075160575) / 2	1.149
Jul	7	1.25917842		
	19	1.127310355	(1.25917842 + 1.127310355) / 2	1.193
Aug	8	1.169483936		
	20	1.101262157	(1.169483936 + 1.101262157) / 2	1.135
Sep	9	1.017987712		
	21	1.041897751	(1.017987712 + 1.041897751) / 2	1.030
Oct	10	0.871443827		
	22	0.985955145	(0.871443827 + 0.985955145) / 2	0.929
Nov	11	0.794070513		
	23	0.965086207	(0.794070513 + 0.965086207) / 2	0.880
Dec	12	0.728028075		
	24	0.942140034	(0.728028075 + 0.942140034) / 2	0.835

Note: These are the raw seasonal indices.

5.4 Adjustment of Seasonal Indices

In a multiplicative model, the 12 monthly seasonal indices should sum to 12. Since the raw indices may not total exactly 12 due to data variation, an adjustment is necessary.

5.4.1 Compute Adjustment Factor

$$\text{Adjustment Factor} = \frac{12}{\text{Sum of Raw Seasonal Indices}} \quad (6)$$

Sum of Raw Seasonal Indices

$$\begin{aligned} \text{Sum} &= 0.796 + 0.846 + 0.894 + 0.939 + 1.021 + 1.149 + 1.193 + 1.135 + 1.030 \\ &\quad + 0.929 + 0.880 + 0.835 \end{aligned}$$

Sum of Raw Seasonal Indices = 11.647

Using equation (6), the Adjustment Factor = $\frac{12}{11.647} = 1.0307$

5.4.2 Adjust Each Monthly Index

$$\text{Adjusted } S_t = \text{Raw } S_t \times \text{Adjustment factor} \quad (7)$$

Table 6: Adjusted seasonal indices

Month	Average SI	Adjusted SI (Calculation)
Jan	0.796	$0.796 \times 1.0307 \approx 0.820$
Feb	0.846	$0.846 \times 1.0307 \approx 0.872$
Mar	0.894	$0.894 \times 1.0307 \approx 0.921$
Apr	0.939	$0.939 \times 1.0307 \approx 0.968$
May	1.021	$1.021 \times 1.0307 \approx 1.053$
Jun	1.149	$1.149 \times 1.0307 \approx 1.184$
Jul	1.193	$1.193 \times 1.0307 \approx 1.230$
Aug	1.135	$1.135 \times 1.0307 \approx 1.170$
Sep	1.030	$1.030 \times 1.0307 \approx 1.062$
Oct	0.929	$0.929 \times 1.0307 \approx 0.958$
Nov	0.880	$0.880 \times 1.0307 \approx 0.907$
Dec	0.835	$0.835 \times 1.0307 \approx 0.861$

Note: Adjusted seasonal indices now sum to 12 for proper normalisation and can be used to deseasonalise the data.

6. Deseasonalisation of the Series

Deseasonalisation removes the seasonal effect from the observed series, leaving the trend and irregular component. This is necessary before fitting a trend equation and forecasting.

Formula (Multiplicative Model):

$$\text{Deseasonalised Value} = \frac{Y_t}{S_t} \quad (8)$$

Where:

- Y_t = observed series value
- S_t = final seasonal index for the corresponding month

Table 7: Deseasonalisation of the series using the multiplicative model

Period	Month	Y_t	S_t	Deseasonalised Value (Y_t / S_t)
6	Jun	21.944	1.184	$21.944 / 1.184 \approx 18.53$
7	Jul	22.499	1.230	$22.499 / 1.230 \approx 18.29$
8	Aug	20.894	1.170	$20.894 / 1.170 \approx 17.85$
9	Sep	18.308	1.062	$18.308 / 1.062 \approx 17.24$
10	Oct	15.913	0.929	$15.913 / 0.929 \approx 17.13$
11	Nov	14.865	0.880	$14.865 / 0.880 \approx 16.88$
12	Dec	14.107	0.835	$14.107 / 0.835 \approx 16.89$
13	Jan	13.662	0.820	$13.662 / 0.820 \approx 16.66$
14	Feb	15.728	0.872	$15.728 / 0.872 \approx 18.03$
15	Mar	18.086	0.921	$18.086 / 0.921 \approx 19.64$
16	Apr	20.696	0.968	$20.696 / 0.968 \approx 21.38$
17	May	24.598	1.021	$24.598 / 1.021 \approx 24.08$
18	Jun	28.624	1.184	$28.624 / 1.184 \approx 24.17$
19	Jul	31.594	1.230	$31.594 / 1.230 \approx 25.69$
20	Aug	32.327	1.170	$32.327 / 1.170 \approx 27.63$
21	Sep	31.843	1.062	$31.843 / 1.062 \approx 29.98$
22	Oct	31.169	0.929	$31.169 / 0.929 \approx 33.55$
23	Nov	31.346	0.880	$31.346 / 0.880 \approx 35.63$
24	Dec	31.231	0.835	$31.231 / 0.835 \approx 37.41$
25	Jan	32.264	0.820	$32.264 / 0.820 \approx 39.33$
26	Feb	33.430	0.872	$33.430 / 0.872 \approx 38.34$
27	Mar	34.344	0.921	$34.344 / 0.921 \approx 37.28$
28	Apr	36.164	0.968	$36.164 / 0.968 \approx 37.36$
29	May	36.164	1.021	$36.164 / 1.021 \approx 35.41$

7. Trend Estimation

After deseasonalising the data in Step 6, the trend component is estimated using the least squares regression method.

7.1 Trend model

We use the linear trend equation:

$$T_t = a + bt \quad (9)$$

Where:

- T_t = trend at time t
- t = period
- a = intercept
- b = slope

7.2 Required formulas

Slope b :

$$b = \frac{n \sum(t \cdot Y_{ds}) - \sum t \sum Y_{ds}}{n \sum t^2 - (\sum t)^2} \quad (10)$$

Intercept a :

$$a = \frac{\sum Y_{ds} - b \sum t}{n} \quad (11)$$

Where:

- Y_{ds} = Deseasonalised value at period t
- n = Number of observations
- t = Time period index

7.3 Calculations

We have deseasonalised values Y_{ds} (from Step 6.2) for period ($t = 6-29$) shown in the table below. Using that we calculate..

The deseasonalised values Y_{ds} obtained in Step 6.2 for periods $t = 6$ to 29 are used to compute the parameters of the linear trend model.

To facilitate the estimation of the trend equation using the least squares method, the following quantities are calculated:

- t : Time index assigned sequentially to the deseasonalised data (from 1 to 24 corresponding to periods 6–29).
- Y_{ds} – Deseasonalised value for each period.
- $t \times Y_{ds}$ – Product of the time index and the deseasonalised value, used to calculate the slope of the trend.
- t^2 – Square of the time index, used in the calculation of the slope and intercept.

Table 8: Computation of trend parameters using deseasonalised values

t	Period	Y_{ds}	$t \times Y_{ds}$ (calculation)	t^2 (calculation)
1	6	18.52	$1 \times 18.52 = 18.52$	$1^2 = 1$
2	7	18.29	$2 \times 18.29 = 36.58$	$2^2 = 4$
3	8	17.85	$3 \times 17.85 = 53.55$	$3^2 = 9$
4	9	17.24	$4 \times 17.24 = 68.96$	$4^2 = 16$
5	10	17.13	$5 \times 17.13 = 85.65$	$5^2 = 25$
6	11	16.88	$6 \times 16.88 = 101.28$	$6^2 = 36$
7	12	16.89	$7 \times 16.89 = 118.23$	$7^2 = 49$
8	13	16.66	$8 \times 16.66 = 133.28$	$8^2 = 64$
9	14	18.03	$9 \times 18.03 = 162.27$	$9^2 = 81$
10	15	19.64	$10 \times 19.64 = 196.40$	$10^2 = 100$
11	16	21.38	$11 \times 21.38 = 235.18$	$11^2 = 121$
12	17	24.08	$12 \times 24.08 = 288.96$	$12^2 = 144$
13	18	24.17	$13 \times 24.17 = 314.21$	$13^2 = 169$
14	19	25.69	$14 \times 25.69 = 359.66$	$14^2 = 196$
15	20	27.63	$15 \times 27.63 = 414.45$	$15^2 = 225$
16	21	29.98	$16 \times 29.98 = 479.68$	$16^2 = 256$
17	22	33.55	$17 \times 33.55 = 570.35$	$17^2 = 289$
18	23	35.63	$18 \times 35.63 = 641.34$	$18^2 = 324$
19	24	37.41	$19 \times 37.41 = 710.79$	$19^2 = 361$
20	25	39.33	$20 \times 39.33 = 786.60$	$20^2 = 400$
21	26	38.34	$21 \times 38.34 = 804.14$	$21^2 = 441$
22	27	37.28	$22 \times 37.28 = 820.16$	$22^2 = 484$
23	28	37.36	$23 \times 37.36 = 858.28$	$23^2 = 529$
24	29	35.41	$24 \times 35.41 = 849.84$	$24^2 = 576$

7.4 Estimation of the Linear Trend Equation

Summing the values from Table 8, we obtain:

$$\Sigma t = 300, \quad \Sigma Y_{ds} = 624.37, \quad \Sigma t \times Y_{ds} = 9108.36 \text{ and } \Sigma t^2 = 4,900$$

Note: Complete step-by-step calculations are provided in the annex (hand calculations).

7.4.1 Compute the Slope b

The slope of the trend line is calculated using the least squares formula:

$$b = \frac{n \Sigma(t \cdot Y_{ds}) - \Sigma t \Sigma Y_{ds}}{n \Sigma t^2 - (\Sigma t)^2}$$

Substituting the values:

$$b = \frac{24 \cdot 9108.36 - 300 \cdot 624.37}{24 \cdot 4900 - (300)^2} = \frac{31289.64}{27600} = 1.13368$$
$$b \approx 1.134$$

7.4.2 Compute the Intercept a

The intercept is calculated as:

$$a = \frac{\Sigma Y_{ds} - b \Sigma t}{n}$$

Substituting the values:

$$a = \frac{624.37 - (1.134 \cdot 300)}{24} = \frac{284.17}{24} = 11.84$$

7.4.3 : Trend Equation

Therefore, the **linear trend equation** is: $T_t = 11.84 + 1.134 t$

Where:

- T_t is the estimated trend value at time t
- t is the time index (1 for period 6, 2 for period 7, ... 24 for period 29)

8. Forecasting Using the Trend Equation and Seasonal Indices

We use the multiplicative model:

$$\text{Forecast} = T_t \times S_t \quad (12)$$

Where:

- $T_t = 11.84 + 1.134 t$ (trend equation)
- S_t = seasonal index for the corresponding month

8.1. Determine t for Forecast Periods

- Last observed period: period 29 $\rightarrow t = 24$
- Forecast periods: 37–44 $\rightarrow$ continue sequentially:

Table 9: Time index (t) for forecast periods 37–44.

Period	t
37	32
38	33
39	34
40	35
41	36
42	37
43	38
44	39

8.2. Adjusted Seasonal Indices

Using adjusted seasonal indices for each month (from earlier calculation):

Table 10: Adjusted seasonal indices (S_t)

Month	Adjusted S_t
Jan	0.820
Feb	0.872
Mar	0.921
Apr	0.968
May	1.053
Jun	1.184
Jul	1.230
Aug	1.170

Sep	1.062
Oct	0.958
Nov	0.907
Dec	0.861

8.2.3. Compute Trend Component

Table 11: Trend values T_t for forecast periods using the linear model.

Period	t	Trend Calculation	T_t (Trend)
37	32	$11.84 + 1.134 \times 32$	48.33
38	33	$11.84 + 1.134 \times 33$	49.47
39	34	$11.84 + 1.134 \times 34$	50.60
40	35	$11.84 + 1.134 \times 35$	51.73
41	36	$11.84 + 1.134 \times 36$	52.86
42	37	$11.84 + 1.134 \times 37$	53.99
43	38	$11.84 + 1.134 \times 38$	55.12
44	39	$11.84 + 1.134 \times 39$	56.25

8.2.4. Compute Forecasts

Assuming the months for periods 37–44 are Jan→ Aug:

Table 12: Forecast values using multiplicative model ($T_t \times S_t$) for periods 37–44.

Period	Month	T_t	S_t	Forecast Calculation ($T_t \times S_t$)	Forecast Y_t
37	Jan	48.33	0.820	48.33×0.820	39.6306
38	Feb	49.47	0.872	49.47×0.872	43.1378
39	Mar	50.60	0.921	50.60×0.921	46.6026
40	Apr	51.73	0.968	51.73×0.968	50.0746
41	May	52.86	1.053	52.86×1.053	55.66158
42	June	53.99	1.184	53.99×1.184	63.92416
43	July	55.12	1.230	55.12×1.230	67.7976
44	Aug	56.25	1.170	56.25×1.170	65.8125

Note: The forecast for each period is obtained by multiplying the trend component, T_t with the seasonal index S_t for the corresponding month using the multiplicative model.

9. Conclusion

This study analysed the time series using a classical decomposition approach. The time series plot revealed a clear upward trend along with a repeating seasonal pattern, justifying the use of a multiplicative decomposition model.

The series was decomposed into trend and seasonal components, and a linear trend was fitted using the deseasonalised data.

Forecasts for periods 37–44 indicate a continued increase in the series, with higher values in months such as June to August due to stronger seasonal effects. Overall, the model effectively captures both trend and seasonality, providing reliable forecasts for future periods.

Question 2: ARIMA Time Series Forecasting

1. Introduction

This report analyses the assigned time series for Student ID 2523258 (Series ID 22), consisting of 80 monthly observations. The main objective is to identify and fit an appropriate ARMA/ARIMA model, perform diagnostic checks, and generate forecasts for the next four months. All R outputs, plots, and interpretations are presented to support the modelling process.

2. Data Preparation

The dataset was imported and filtered to extract the relevant series using the student ID. The observations were then ordered by time period and converted into a time series object with a frequency of 12, assuming monthly data.

3. Time Series Visualisation

Figure 2 displays the monthly observations of the time series over 80 periods.

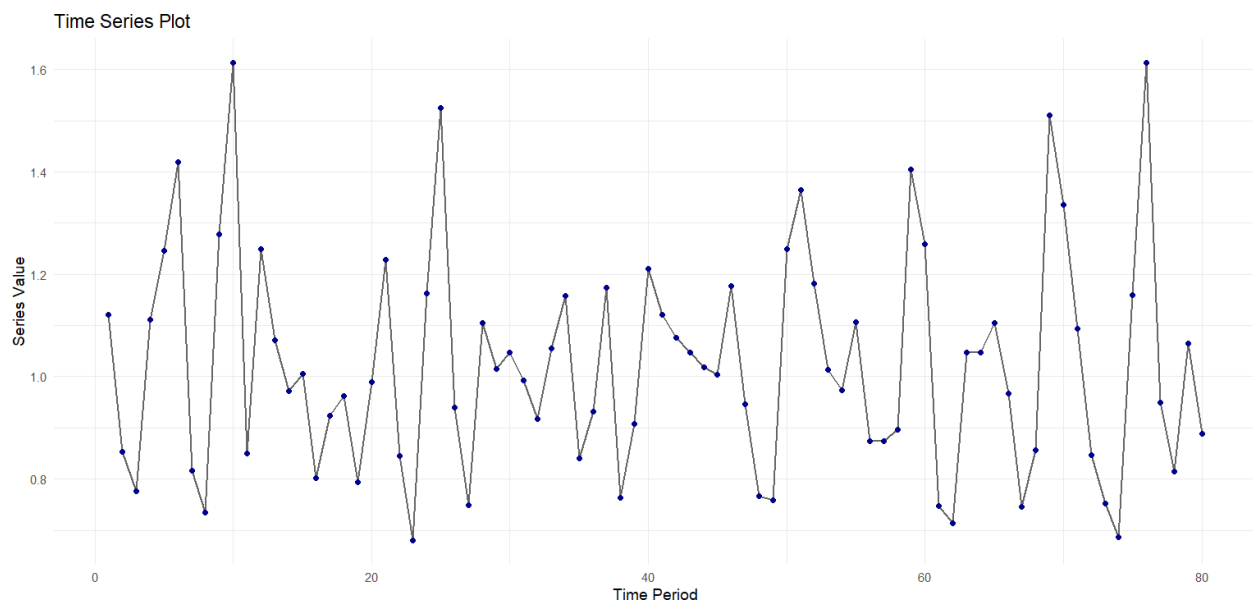


Figure 2: Original monthly time series for Student ID 2523258. The gray line shows the overall trend, and the blue points represent the observed values.

Analysis:

The series fluctuate around a relatively stable mean ranging approximately from 0.7 to 1.6.

There is no clear long-term trend, and the changes appear mostly random, with occasional peaks and troughs. While there are some periodic spikes, no strong seasonal pattern is visible, and the variance remains stable throughout the period.

The plot suggests that the series is likely stationary since the mean and variance do not change systematically over time. However, this needs to be confirmed with a formal test, which is discussed in the next section.

4. Stationarity Test (ADF Test)

R Output:

Table 13: ADF test results.

Statistic	Lag Order	p-value	Alternative Hypothesis
-5.1686	4	0.01	stationary

Interpretation:

The ADF test gives a p-value of 0.01, which is less than the 0.05 significance level. This indicates that the series is stationary, meaning it does not have a unit root. The result supports the initial visual impression from Figure 2, and no differencing is required before modelling.

5. ARIMA Model Identification

5.1 ACF and PACF Plots

Figure 3 shows the Autocorrelation Function (ACF) in green, while Figure 4 shows the Partial Autocorrelation Function (PACF) in orange. These plots are used to identify the potential orders of the autoregressive (AR) and moving average (MA) components for the ARIMA model.

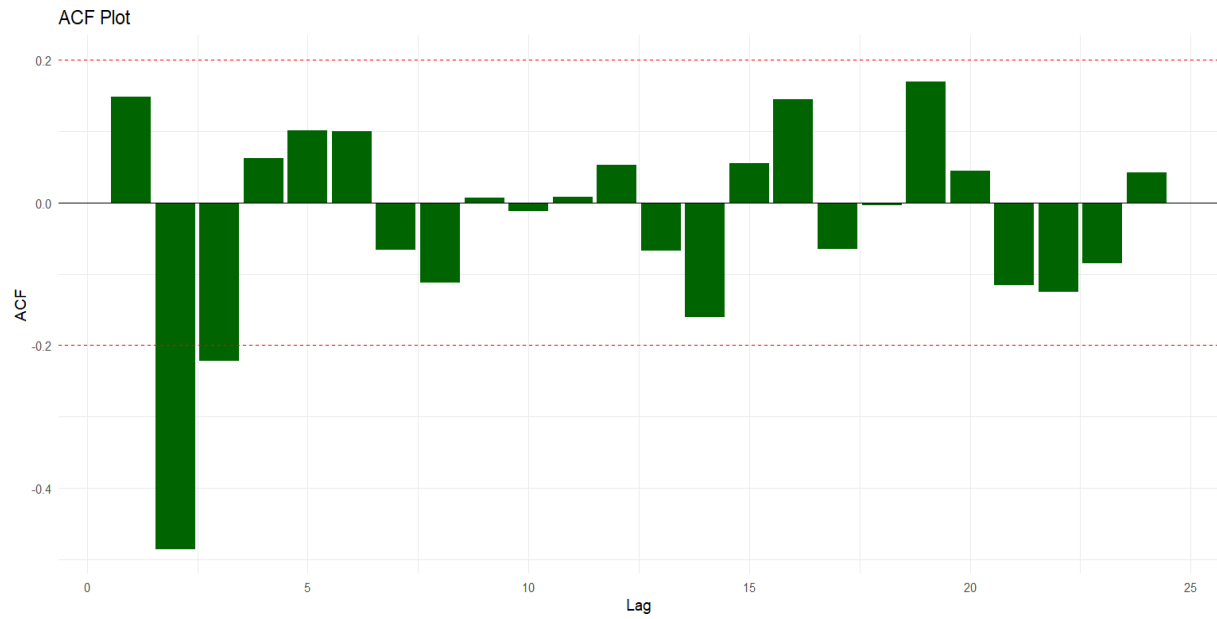


Figure 3: ACF plot

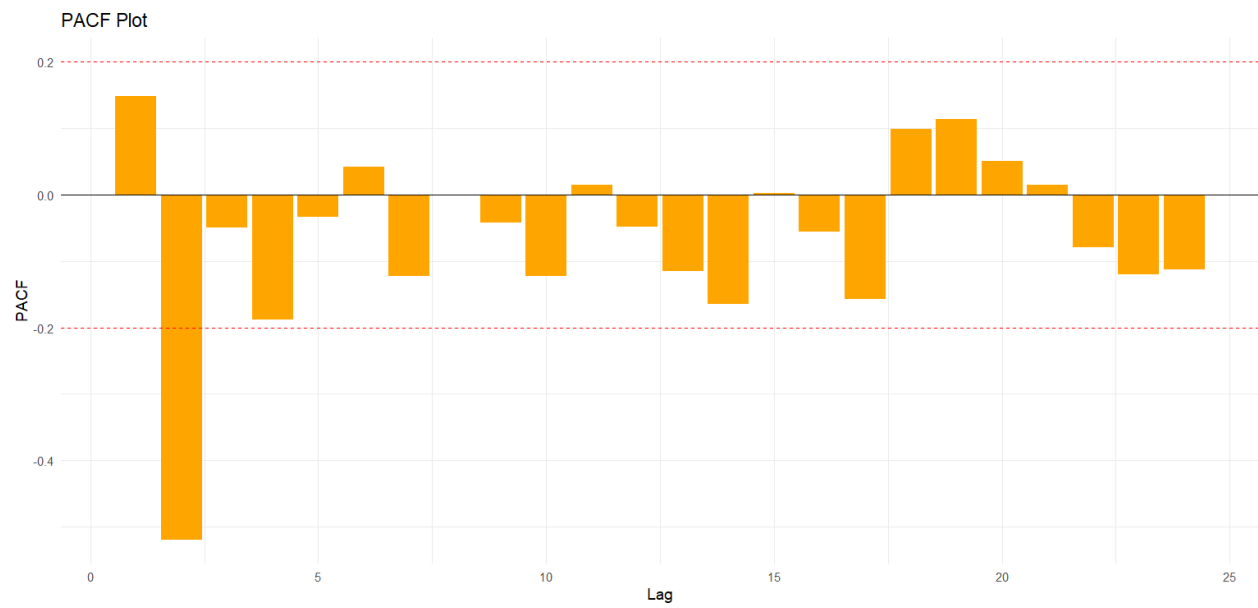


Figure 4: PACF plot

ACF Plot Interpretation:

The ACF plot shows correlations between the series and its lagged values. Most of the bars fall within the significance bounds (red dashed lines), indicating that correlations at higher lags are not statistically significant. The first few lags show a small positive correlation, while some early negative spikes are present (e.g., lag 2). Overall, the ACF decays relatively quickly, suggesting that the series does not have strong long-term autocorrelation. This pattern indicates that an MA component with a small number of terms (e.g., 1 or 2) may be sufficient to model the short-term dependencies in the series.

PACF Plot Interpretation:

The PACF plot isolates the direct correlation between the series and its lags, removing the effects of intervening lags. The first lag shows a small positive correlation, while the second lag shows a negative correlation. Beyond lag 2, the PACF values mostly remain within the significance bounds, with minor fluctuations at later lags. This pattern suggests that including two autoregressive terms (AR(2)) is likely sufficient, as additional AR terms would not contribute significantly to explaining the series' behaviour.

Conclusion from ACF & PACF

Together, these plots suggest that the series contains short-term dependencies that can be captured using a combination of AR and MA terms. This supports the selection of an **ARIMA(2,0,2)** model.

5.2 Fitted ARIMA Model

Based on the analysis, an ARIMA(2,0,2) model with a non-zero mean was fitted. The parameter estimates are summarised below:

Table 14: Parameter estimates for the fitted ARIMA(2,0,2) model with non-zero mean.

Parameter	Estimate	Std. Error	Interpretation
ar1	0.3907	0.1917	Positive correlation with the previous period
ar2	-0.3141	0.2132	Negative correlation with the period two steps back
ma1	-0.2189	0.2009	Short-term shock adjustment
ma2	-0.3242	0.2250	Additional short-term error correlation
mean	1.0248	0.0102	Overall average level of the series

Model fit statistics, including AIC and BIC, indicate a reasonable fit. The residuals show minimal autocorrelation, suggesting that the model adequately captures the series dynamics.

Model Fit Statistics:

Table 15: Model Fit Statistics

Metric	Value
σ^2	0.03393
Log Likelihood	23.99
AIC	-35.99
AICc	-34.84
BIC	-21.7
RMSE	0.1783
MAE	0.1396
MAPE	14.04%
ACF1 Residual	0.0076

Interpretation:

The relatively low AIC and BIC values indicate a good model fit. The residual autocorrelation ($ACF1 \approx 0$) is close to zero, suggesting that the model has adequately captured the underlying structure of the series.

5.3 Model Comparison

Alternative models were also evaluated:

Table 16: Model Comparison

Model	AIC
ARIMA(1,0,1)	-22.21
ARIMA(2,0,1)	-35.58
ARIMA(1,0,2)	-39.08

Although the ARIMA(1,0,2) model has a slightly lower AIC, the **ARIMA(2,0,2)** model was selected as it provides a more balanced and robust representation of both autoregressive and moving average components.

6. Residual Diagnostics

Figure 5 provides a comprehensive look at the residuals from the fitted model. The top panel plots residual values over time, showing fluctuations centered around zero with no discernible patterns or trends. The two lower panels display the autocorrelation function (ACF) and partial autocorrelation function (PACF) of the residuals, respectively. The absence of significant spikes beyond the confidence bounds in these plots suggests that there is no remaining autocorrelation in the residuals. This indicates that the model has successfully captured the temporal dependencies in the data, and the residuals behave like white noise, an essential condition validating the adequacy of the ARIMA model.

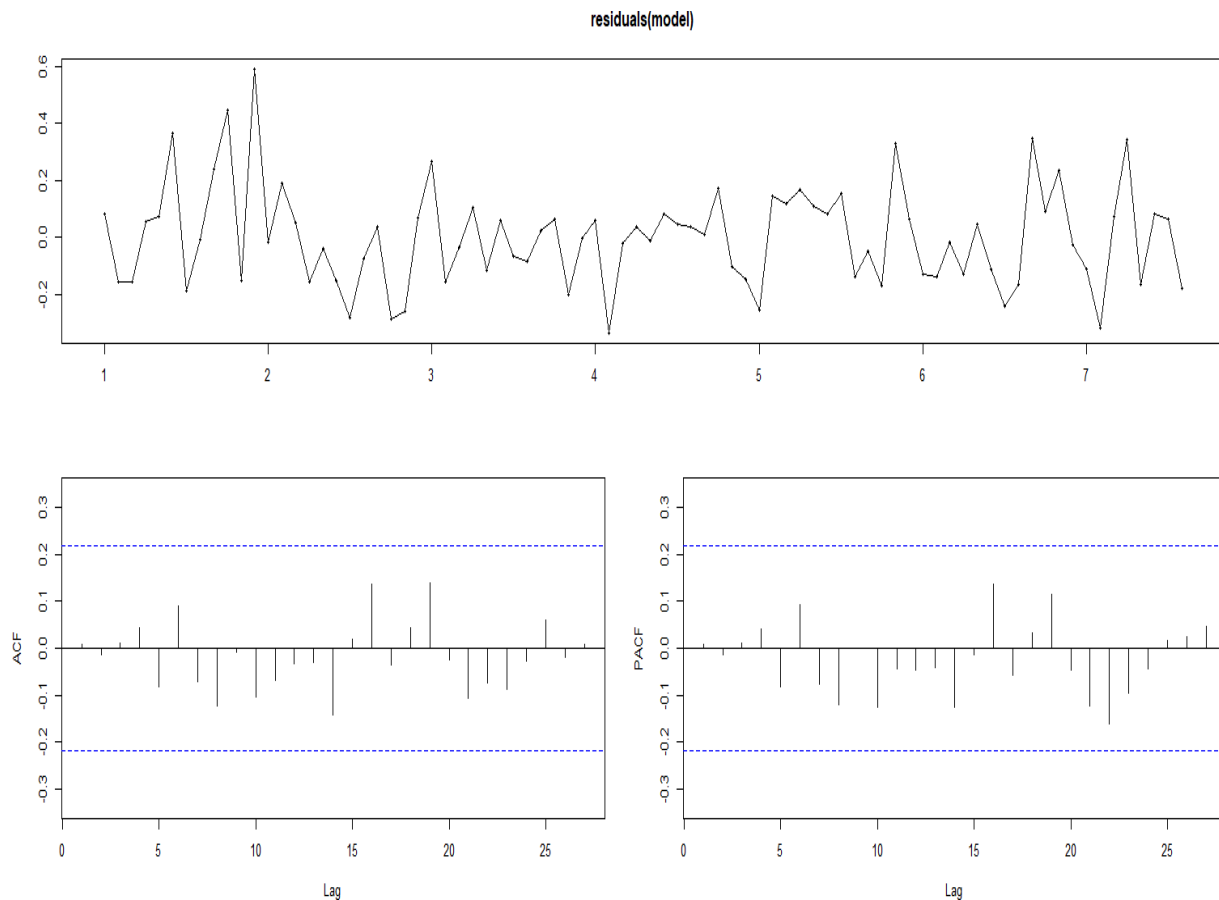


Figure 5: Standard Residual Diagnostics (Residuals over Time and ACF/PACF)

Figure 6 presents a histogram and density curve of the residuals alongside a time series plot of the residuals.

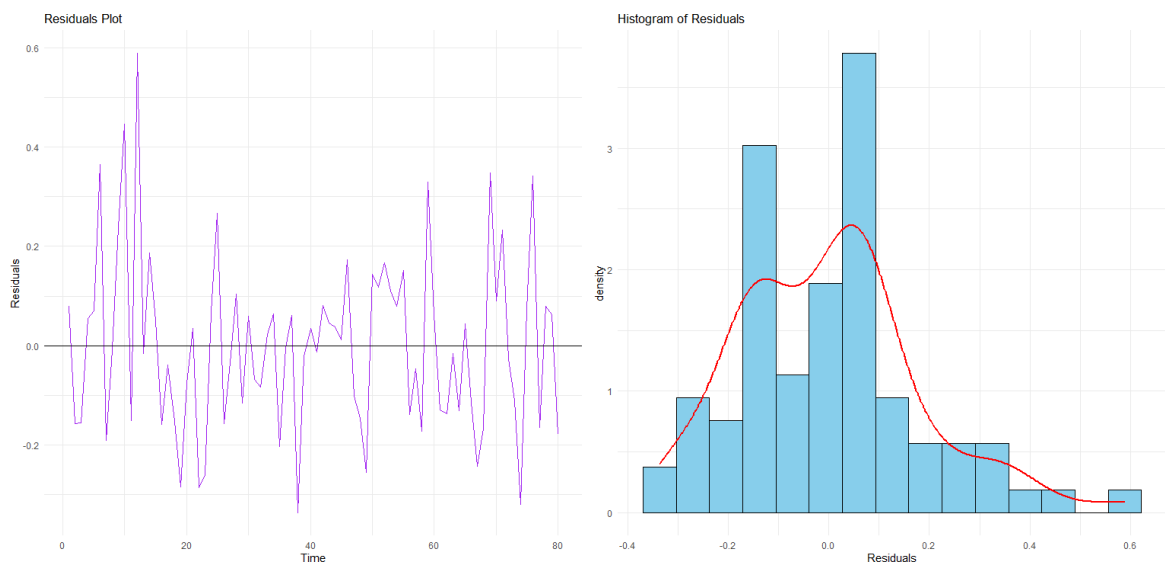


Figure 6: Histogram, Density, and Time Series Plot of Residuals

The histogram (sky blue) with the red density curve shows the residuals are approximately normally distributed without noticeable skewness or extreme outliers. The time series plot (purple line) reveals random fluctuations over time without any discernible patterns. Together, these plots indicate that the residuals are normally distributed and independent, further validating the model’s adequacy.

Box-Ljung Test Results

Table 17: Box-Ljung Test Results

Statistic	Degrees of Freedom (df)	p-value
4.2647	10	0.9346

The Box-Ljung test was performed on the residuals of the ARIMA(2,0,2) model to detect any remaining autocorrelation. The test statistic is 4.2647 with 10 degrees of freedom, yielding a p-value of 0.9346. Since the p-value is substantially greater than the 0.05 significance level, we fail to reject the null hypothesis of no autocorrelation. This indicates that the residuals behave like white noise and that the model has successfully captured the dependencies in the data.

Together with the residual diagnostics, these results confirm that the ARIMA(2,0,2) model provides an appropriate and reliable representation of the time series

7. Forecasting Next 4 Months

The ARIMA(2,0,2) model was used to forecast the next four months as shown in Figure 7.

Table 18: Forecasts for the next four months using the ARIMA(2,0,2) model with confidence intervals.

Period	Point Forecast	80% CI	95% CI
Sep 7	0.978	(0.742,1.214)	(0.617,1.339)
Oct 7	1.107	(0.868,1.347)	(0.741,1.474)
Nov 7	1.072	(0.797,1.347)	(0.651,1.492)
Dec 7	1.017	(0.735,1.300)	(0.585,1.449)

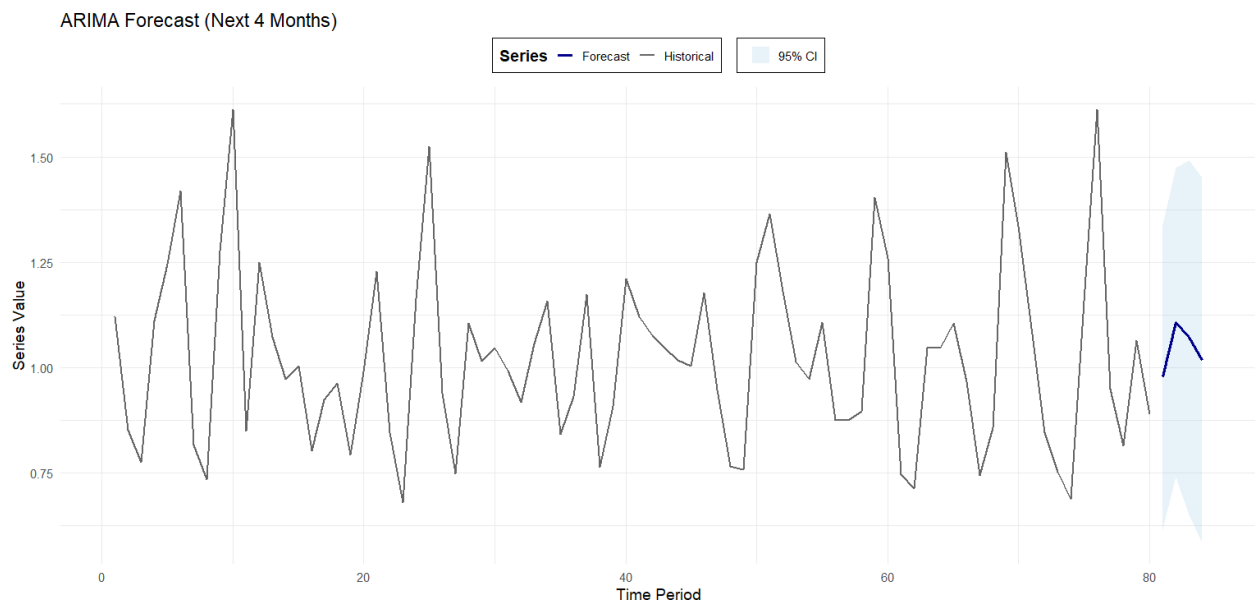


Figure 7: Forecast for the next 4 months with ARIMA(2,0,2). Gray line: historical series; blue line: forecasted values; light blue shaded area: 95% confidence interval

Interpretation:

The point forecasts range from 0.978 to 1.107, which is consistent with the historical series. The 80% confidence intervals suggest moderate certainty, while the 95% intervals are wider to reflect uncertainty in the forecast.

The forecasts are stable and align with past behaviour, providing a reasonable expectation for the short-term future. This demonstrates that the model captures both the level and short-term variation of the series.

8. Conclusion

The time series was found to be stationary, eliminating the need for differencing. An ARIMA(2,0,2) model with a non-zero mean was identified as appropriate based on ACF/PACF analysis and model selection criteria.

Diagnostic tests confirmed that the residuals behave like white noise and are approximately normally distributed, validating the model assumptions. The generated forecasts are stable and consistent with historical patterns, demonstrating that the model provides reliable short-term predictions.

ANNEX

Question 1

A.1 R Code for Time Series Plot (Step 2)

The following R code was used to generate the time series plot for preliminary analysis of the data, as described in Step 2 of the report.

```
# Data Modelling Assignment

# Student ID: 2523258

# Question 1
# Time Series Decomposition and Forecasting Model

# 1. Load Required Libraries

library(dplyr)
library(ggplot2)
library(forecast)
library(scales)
library(gridExtra)

# 2. Load Dataset

setwd("C:/Users/gnasr/OneDrive/Desktop/DM 2 Assignment")

data <- read.csv("Decomp_series_data-Data Modelling Assignment.csv")

# 3. Extract Series for Student ID

student_id <- "2523258"

my_data <- data %>%
  filter(grepl(student_id, Series.Name)) %>%
  arrange(Time.Period)

# 4. Create Monthly Time Series Object

ts_data <- ts(
  my_data$Series.Value,
  frequency = 12
)
```

```

# 5. Time Series Plot

ts_plot <- ggplot(my_data, aes(x = Time.Period, y = Series.Value)) +

  geom_line(color = "navy", size = 1.4, show.legend = FALSE) +
  geom_point(color = "navy", size = 2.5, show.legend = FALSE) +

  # Clean axis numbers
  scale_x_continuous(breaks = pretty_breaks(n = 10)) +
  scale_y_continuous(breaks = pretty_breaks(n = 8)) +

  # Labels
  labs(
    title = "Monthly Time Series",
    subtitle = "Student ID: 2523258",
    x = "Time Period",
    y = "Series Value"
  ) +

  # Theme
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 18),
    plot.subtitle = element_text(size = 13),
    axis.title = element_text(face = "bold")
  )

print(ts_plot)

```

A.2 Hand Calculations for Moving Averages (Step 4.2)

The following shows the detailed step-by-step hand calculations used to obtain the moving averages (MA) for each period (Step 4.2)

Compute 12-Month Moving Average (MA).

We compute MA as the average of 12 consecutive months using equation (2):

$$MA_t = \frac{Y_t + Y_{t-1} + \dots + Y_{t-11}}{12} \quad \text{where } Y_t = \text{observed value at time } t$$

MA_t = 12-month moving average at time t

• MA_6 (periods 1-12)

$$MA_6 = \frac{14.889 + 16.347 + 17.513 + 18.428 + 20.248 + 21.944 + 22.494 + 20.894 + 18.308 + 15.913 + 14.865 + 14.107}{12}$$

$$= \frac{215.955}{12}$$

$$= 17.99625$$

$$\approx 17.996$$

• MA_7 (periods 2-12)

$$MA_7 = \frac{16.347 + 17.513 + 18.428 + 20.248 + 21.944 + 22.499 + 20.894 + 18.308 + 15.913 + 14.865 + 14.107 + 13.662}{12}$$

$$= \frac{214.728}{12}$$

$$= 17.894$$

- MA_8 (periods 3-14)

$$MA_8 = \frac{17.513 + 18.423 + 20.248 + 21.944 + 22.499 + 20.894 + 18.308 + 15.913 + 14.865 + 14.107 + 13.662 + 15.728}{12}$$

$$= \frac{214.109}{12}$$

$$= 17.842$$

- MA_9 (periods 4-15)

$$MA_9 = \frac{18.428 + 20.248 + 21.944 + 22.499 + 20.894 + 18.308 + 15.913 + 14.865 + 14.107 + 13.662 + 15.728 + 18.086}{12}$$

$$= \frac{214.682}{12}$$

$$= 17.890$$

- MA_{10} (periods 5-16)

$$MA_{10} = \frac{20.248 + 21.944 + 22.499 + 20.894 + 18.308 + 15.913 + 14.865 + 14.107 + 13.662 + 15.728 + 18.086 + 20.696}{12}$$

$$= \frac{216.950}{12} = 18.079$$

• MA_{11} (periods 6-17)

$$MA_{11} = 81.944 + 22.499 + 20.894 + 18.308 + 15.913 + 14.865 + 14.107 \\ + 13.662 + 15.728 + 18.086 + 20.696 + 24.598$$

12

$$= \frac{221.300}{12}$$

$$= 18.422$$

• MA_{12} (periods 7-18)

$$MA_{12} = 22.499 + 20.894 + 18.308 + 15.913 + 14.865 + 14.107 + 13.662 \\ + 15.728 + 18.086 + 20.696 + 24.598 + 28.624$$

12

$$= \frac{227.980}{12}$$

$$= 18.998$$

• MA_{13} (periods 8-19)

$$MA_{13} = 20.894 + 18.308 + 15.913 + 14.865 + 14.107 + 13.662 + 15.728 \\ + 18.086 + 20.696 + 24.598 + 28.624 + 31.594$$

12

$$= \frac{237.075}{12}$$

$$= 19.756$$

- MA_{14} (periods 9-20)

$$MA_{14} = 18.308 + 15.913 + 14.865 + 14.107 + 13.662 + 15.728 \\ + 18.086 + 20.696 + 24.598 + 28.624 + 31.594 + 32.327$$

12

$$= \frac{248.508}{12}$$

$$= 20.709$$

- MA_{15} (periods 10-21)

$$MA_{15} = 15.913 + 14.865 + 14.107 + 13.662 + 15.728 + \\ 18.086 + 20.696 + 24.598 + 28.624 + 31.594 + 32.327 \\ + 31.843$$

12

$$= \frac{262.043}{12}$$

$$= 21.837$$

- MA_{16} (periods 11-22)

$$MA_{16} = 14.865 + 14.107 + 13.662 + 15.728 + 18.086 + 20.696 + \\ 24.598 + 28.624 + 31.594 + 32.327 + 31.843 + 31.169$$

12

$$= \frac{277.299}{12} = 23.108$$

• MA_{17} (periods 12-23)

$$MA_{17} = 14.107 + 13.662 + 15.728 + 18.086 + 20.696 + 24.598 + 28.624 + 31.594 + 32.327 + 31.843 + 31.169 + 31.346$$

12

$$= \frac{293.780}{12}$$

$$= 24.482$$

• MA_{18} (periods 13-24)

$$MA_{18} = 13.662 + 15.728 + 18.086 + 20.696 + 24.598 + 28.624 + 31.594 + 32.327 + 31.843 + 31.169 + 31.346 + 31.231$$

12

$$= \frac{310.904}{12}$$

$$= 25.909$$

• MA_{19} (periods 14-25)

$$MA_{19} = 15.728 + 18.086 + 20.696 + 24.598 + 28.624 + 31.594 + 32.327 + 31.843 + 31.169 + 31.346 + 31.231 + 30.805$$

12

$$= \frac{328.047}{12}$$

$$= 27.337$$

• MA_{20} (periods 15-20)

$$MA_{20} = 18.086 + 20.696 + 24.598 + 28.624 + 31.594 + 32.327 \\ + 31.843 + 31.169 + 31.346 + 31.231 + 30.805 + 32.264$$

12

$$= \frac{344.583}{12}$$

$$= 28.715$$

• MA_{21} (periods 16-27)

$$MA_{21} = 20.696 + 24.598 + 28.624 + 31.594 + 32.327 + 31.843 \\ + 31.169 + 31.346 + 31.231 + 30.805 + 32.264 + 33.430$$

12

$$= \frac{359.427}{12}$$

$$= 29.944$$

• MA_{22} (periods 17-28)

$$MA_{22} = 24.598 + 28.624 + 31.594 + 32.327 + 31.843 + 31.169 + 31.346 \\ + 31.231 + 30.805 + 32.264 + 33.430 + 34.344$$

12

$$= \frac{373.575}{12}$$

$$= 31.131$$

• MA_{23} (periods 18-29)

$$MA_{23} = 28.624 + 31.594 + 32.327 + 31.843 + 31.169 + 31.346 + 31.231 + 30.805 + 32.264 + 32.430 + 34.344 + 36.164$$

12

$$= \frac{385.141}{12}$$

$$= 32.095$$

• MA_{24} (periods 19-30)

$$MA_{24} = 31.594 + 32.327 + 31.843 + 31.169 + 31.346 + 31.231 + 30.805 + 32.264 + 32.430 + 34.344 + 36.164 + 37.861$$

12

$$= \frac{394.378}{12}$$

$$= 32.865$$

• MA_{25} (periods 20-31)

$$MA_{25} = 32.327 + 31.843 + 31.169 + 31.346 + 31.231 + 30.805 + 32.264 + 32.430 + 34.344 + 36.164 + 37.861 + 38.415$$

12

$$= \frac{401.199}{12}$$

$$= 33.433$$

• MA_{26} (periods 21-32)

$$MA_{26} = \frac{31.843 + 31.169 + 31.346 + 31.231 + 30.805 + 32.264 + 33.430 + 34.344 + 36.164 + 37.861 + 38.415 + 36.810}{12}$$

$$= \frac{405.682}{12}$$

$$= 33.807$$

• MA_{27} (periods 22-33)

$$MA_{27} = \frac{31.169 + 31.346 + 31.231 + 30.805 + 32.264 + 33.430 + 34.344 + 36.164 + 37.861 + 38.415 + 36.810 + 34.225}{12}$$

$$= \frac{408.064}{12}$$

$$= 34.005$$

• MA_{28} (periods 23-34)

$$MA_{28} = \frac{31.346 + 31.231 + 30.805 + 32.264 + 33.430 + 34.344 + 36.164 + 37.861 + 38.415 + 36.810 + 34.225 + 31.829}{12}$$

$$= \frac{408.724}{12}$$

$$= 34.060$$

- MA_{29} (periods 24-35)

$$MA_{29} = \frac{31.231 + 30.805 + 32.264 + 32.430 + 34.344 + 36.164 + 37.861 + 38.415 + 36.810 + 34.225 + 31.829 + 30.781}{12}$$

$$= \frac{408.159}{12}$$

$$= 34.013$$

- MA_{30} (periods 25-36)

$$MA_{30} = \frac{30.805 + 32.264 + 32.430 + 34.344 + 36.164 + 37.861 + 38.415 + 36.810 + 34.225 + 31.829 + 30.781 + 30.024}{12}$$

$$= \frac{406.952}{12}$$

$$= 33.913$$

A.3 Hand Calculations for Summations (Step 7.4)

The following shows the detailed step-by-step hand calculations used to obtain the summations required for estimating the linear trend equation (Step 7.4).

Summation of Required Quantities

From Table 8, the following totals are computed:

Summations

$$\begin{aligned}\sum t &= 1+2+3+4+5+6+7+8+9+10+11+12 \\ &\quad +13+14+15+16+17+18+19+20+21+22+23+24 \\ &= 300 \leftarrow\end{aligned}$$
$$\begin{aligned}\sum Y_{ds} &= 18.52+18.29+17.85+17.24+17.13+16.88 \\ &\quad +16.89+16.66+18.03+19.64+21.38+24.08 \\ &\quad +24.17+25.69+27.63+29.98+33.55+35.63 \\ &\quad +37.41+39.33+38.34+37.28+37.36+35.41 \\ &= 624.37 \leftarrow\end{aligned}$$
$$\begin{aligned}\sum (tY_{ds}) &= 18.52+36.58+53.55+68.96+85.65 \\ &\quad +101.28+118.23+133.28+162.27+ \\ &\quad 196.40+235.18+288.96+314.28 \\ &\quad +359.66+414.45+479.68+570.35 \\ &\quad +641.34+710.79+786.60+804.14 \\ &\quad +820.16+858.28+849.84 \\ &= 9108.36 \leftarrow\end{aligned}$$
$$\begin{aligned}\sum t^2 &= 1+4+9+16+25+36+49+64+81+100+121+144+ \\ &\quad 169+196+225+269+324+361+400+441+484+ \\ &\quad 529+576 = 4900 \leftarrow\end{aligned}$$

Question 2

B.1 R Code for ARIMA Time Series Forecasting (Student ID 2523258)

```
# Data Modelling Assignment
# Question 2: ARIMA Time Series Forecasting
# Student ID: 2523258

# 1. Introduction
# This script implements a complete ARIMA modelling workflow, including
# identification, estimation, diagnostic checking, and forecasting.

# 2. Data Preparation
# Load required libraries, import the dataset,
# and extract the relevant time series.

library(dplyr)
library(ggplot2)
library(forecast)
library(tseries)
library(gridExtra)

# Set working directory
setwd("C:/Users/gnasr/OneDrive/Desktop/DM 2 Assignment")

# Read dataset
raw_data <- read.csv("Arima_series_data-Data Modelling Assignment Data.csv")

# Inspect structure and variable names
head(raw_data)
names(raw_data)

# Filter observations corresponding to the selected student ID: 2523258
id_value <- "2523258"

series_data <- raw_data %>%
  filter(grepl(id_value, Series.name.Student.ID)) %>%
  arrange(Time.Period)

# Convert extracted values into a monthly time series object
ts_data <- ts(series_data$Series.Value, frequency = 12)
```

```

# 3. Time Series Visualisation
# Explore the behaviour of the series over time
# to assess trend, seasonality, and variability.

plot_df <- data.frame(Time = 1:length(ts_data), Value = as.numeric(ts_data))

ggplot(plot_df, aes(x = Time, y = Value)) +
  geom_line(color = "gray40", size = 1) +
  geom_point(color = "blue4", size = 2) +
  ggtitle("Time Series Plot") +
  xlab("Time Period") +
  ylab("Series Value") +
  theme_minimal(base_size = 14)

# 4. Stationarity Test (ADF Test)
# Formally test whether the series is stationary
# and determine if differencing is required.

adf_output <- adf.test(ts_data)
print(adf_output)

# 5. ARIMA Model Identification
# Use ACF and PACF patterns to guide the
# selection of AR and MA components.

acf_vals <- Acf(ts_data, plot = FALSE)
pacf_vals <- Pacf(ts_data, plot = FALSE)

# ACF plot (identifies potential MA terms)
acf_data <- data.frame(Lag = acf_vals$lag[-1], ACF = acf_vals$acf[-1])
ggplot(acf_data, aes(x = Lag, y = ACF)) +
  geom_bar(stat = "identity", fill = "darkgreen") +
  geom_hline(yintercept = 0, color = "black") +
  geom_hline(yintercept = c(0.2, -0.2), linetype = "dashed", color = "red") +
  ggtitle("ACF Plot") +
  theme_minimal(base_size = 14)

```

```

# PACF plot (identifies potential AR terms)
pacf_data <- data.frame(Lag = pacf_vals$lag, PACF = pacf_vals$acf)
ggplot(pacf_data, aes(x = Lag, y = PACF)) +
  geom_bar(stat = "identity", fill = "orange") +
  geom_hline(yintercept = 0, color = "black") +
  geom_hline(yintercept = c(0.2, -0.2), linetype = "dashed", color = "red") +
  ggtitle("PACF Plot") +
  theme_minimal(base_size = 14)

# 6. Model Estimation and Diagnostics
# Fit the ARIMA model, compare alternatives,
# and validate assumptions using residual analysis.

# Fit ARIMA model using automatic selection
arima_model <- auto.arima(ts_data)
summary(arima_model)
coef(arima_model)

# Estimate alternative candidate models for comparison
alt_model1 <- arima(ts_data, order=c(1,0,1))
alt_model2 <- arima(ts_data, order=c(2,0,1))
alt_model3 <- arima(ts_data, order=c(1,0,2))
AIC(alt_model1, alt_model2, alt_model3)

# Extract residuals for diagnostic checking
model_residuals <- residuals(arima_model)
residual_df <- data.frame(Time = 1:length(model_residuals), Residuals = model_residuals)

# Residual time plot (checks randomness)
p1 <- ggplot(residual_df, aes(x = Time, y = Residuals)) +
  geom_line(color = "purple", size = 1) +
  geom_hline(yintercept = 0, color = "black") +
  ggtitle("Residual Plot") +
  theme_minimal(base_size = 14)

# Residual distribution (normality check)
p2 <- ggplot(residual_df, aes(x = Residuals)) +
  geom_histogram(aes(y = ..density..), bins = 15, fill = "skyblue", color = "black") +
  geom_density(color = "red", size = 1) +
  ggtitle("Residual Histogram") +
  theme_minimal(base_size = 14)

```

```

grid.arrange(p1, p2, ncol=2)

# Ljung-Box test to verify absence of autocorrelation
Box.test(model_residuals, lag=10, type="Ljung-Box")

# Additional residual diagnostics (ACF/PACF of residuals)
tsdisplay(residuals(arima_model))

# Evaluate model performance using accuracy metrics
accuracy(arima_model)

# 7. Forecasting
# Generate forecasts and visualise expected future values.

fc_results <- forecast(arima_model, h=4)
print(fc_results)

# Prepare forecast data for plotting
fc_df <- data.frame(
  Time = 81:84,
  Forecast = as.numeric(fc_results$mean),
  Lower95 = fc_results$lower[,2],
  Upper95 = fc_results$upper[,2]
)

# Historical data for comparison
history_df <- data.frame(
  Time = 1:80,
  Value = as.numeric(ts_data)
)

# Forecast plot with confidence intervals
ggplot() +
  geom_line(data = history_df, aes(x = Time, y = Value, color = "Historical"), size = 1) +
  geom_ribbon(data = fc_df, aes(x = Time, ymin = Lower95, ymax = Upper95, fill = "95% CI"), alpha = 0.3) +
  geom_line(data = fc_df, aes(x = Time, y = Forecast, color = "Forecast"), size = 1.2) +
  scale_color_manual(name = "Series", values = c("Historical" = "gray40", "Forecast" = "blue4")) +
  scale_fill_manual(name = "", values = c("95% CI" = "lightblue")) +
  ggtitle("ARIMA Forecast (Next 4 Months)") +
  xlab("Time Period") +
  ylab("Series Value") +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "top",

```

```

    legend.background = element_rect(fill = "white", color = "black"),
    legend.title = element_text(face = "bold")
)

# Create forecast summary table
fc_table <- data.frame(
  Period = 81:84,
  Forecast = as.numeric(fc_results$mean),
  Lower95 = fc_results$lower[,2],
  Upper95 = fc_results$upper[,2]
)
print(fc_table)

# 8. Final Output
# Extract specific point forecasts
fc_results$mean

```